Low-Frequency RoPE Subspace for Near–Far Sparse Attention

Anonymous authors

***Abstract*—Sparse attention must decide which history remains accessible before evaluating attention on a bounded token set. We study the trade-off between candidate access and selection cost at a fixed final token budget. Our near–far cascade combines two mechanisms: a shared 32-coordinate RoPE scoring map for page screening and token refinement, and separate regional page quotas. Final attention uses original-dimensional queries, keys and values. On five Qwen3-8B LongBench tasks, an archived 128/128-protection configuration scores 47.588, compared with 47.910 and 47.666 for nominal full-dimensional page- and token-scoring variants. These descriptive comparisons do not isolate scoring dimensionality or validate the current compact 4/128-protection candidate. In a separate synthetic study, the shared-subspace selector has no lower reported time than the nominal full-dimensional page-scoring variant; historical synchronized TPOT is also higher than dense attention. We therefore present a design and preliminary evaluation, not an established quality–latency improvement. The evidence identifies what a matched evaluation must resolve: whether regional access and compact scoring retain useful tokens at a preparation-inclusive cost below the attention work they remove.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context sparse attention replaces full-history attention with attention over selected tokens. Selection itself, however, requires scoring, ranking and memory access. The central design question is therefore how to preserve access to useful history at a fixed final token budget without spending the saved attention cost on retrieval. A page-to-token cascade makes this tension explicit: page screening reduces the candidate support for refinement, but a useful token cannot be recovered once its page is excluded.

Figure 1 connects these design choices to four Qwen3-8B diagnostics, retaining the token/page perspective of ClusterKV [7]. In the fixed example, mass-ranked, mean-score and extrema-score selection retain different pages (a), motivating a comparison of regional scorers rather than proving a pooling-only effect. Under feasible whole-bin budgets, reserving regional capacity changes distant-history access (b): mean far retention rises from .6759 to .7207, but falls on 43 of 128 correlated observations. Regional allocation is thus a trade-off, not a per-query improvement guarantee. The coordinate diagnostic (c) motivates compact scoring, without isolating frequency from RoPE-pair structure. Historical layer profiles (d) motivate examining selective far computation, but their single-query proxy is not the current gate statistic and provides no decode guarantee. These observations define hypotheses for the cascade; they do not establish production-selector fidelity or task-quality gains.

We examine two coupled decisions within this cascade. The scoring representation determines the information available to rank pages and tokens; regional allocation determines which parts of history can reach refinement. Reducing scoring width does not ensure that distant candidates survive page screening, while reserving distant pages does not ensure inexpensive selection. Our design combines a common low-frequency RoPE coordinate map across both stages with separate near–far page quotas. It retains original-dimensional Q/K/V for attention over the final selected set. The shared map and the regional quotas are the two core mechanisms; a prefill-informed far gate is an optional extension, disabled by default.

This work specifies a shared-map regional cascade and examines three questions: (Q1) what do the available task scores show about compact scoring, (Q2) how do regional budgets constrain candidate access, and (Q3) how do the reported selector and decode costs compare? The evaluation separates historical configuration scores, regional diagnostics and timing measurements rather than combining them into a matched quality–latency result. It reports no established joint advantage: the current compact 4/128-protection candidate lacks matched validation, the synthetic selector time is not lower than the nominal full-dimensional page-scoring variant, and historical synchronized sparse TPOT exceeds dense TPOT. These limits define the status of the design as a research candidate.

II. Related Work

Query-aware retrieval methods such as Quest, SparQ and HISA reduce the active attention set through approximate selection [1], [2], [3]. AsyncTLS provides a page-to-token cascade for sparse selection [4]. We ask whether a shared low-frequency RoPE scoring view and separate near–far page quotas can retain useful candidates at a fixed final token budget. This separates scoring representation and regional allocation from execution scheduling; their quality and preparation-inclusive cost require matched evaluation.

KV compression and placement methods address complementary memory-capacity and data-movement constraints [5], [6], [7]. The method studied here preserves original keys and values and changes their selection. Consequently, it does not inherit a reduction in original-KV cache capacity from the compact scoring representation.

III. Near–Far Selection in a Shared Subspace

The cascade follows one data path (Fig. 2). First, a fixed coordinate map supplies the scoring representations for both stages. L1 selects near and far pages under separate quotas; L2 ranks tokens using that same map and the retained page support. The resulting indices are merged with protected positions, and each query recomputes attention from original-dimensional Q/K/V. This separation keeps approximate scoring out of the retained logits, while candidate exclusion remains an approximation. Ordinary L2 still computes full-history scores before masking, so L1 controls access rather than eliminating those dot products.

A. Shared low-frequency coordinates

For Qwen3-8B, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. They form 16 complete low-frequency rotation pairs, not a contiguous suffix or a learned projection (Fig. 2a). With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. The same map serves both regions and stages within each layer and KV head, while query heads retain their own scores. Slower positional phase variation motivates this choice; it does not guarantee retrieval fidelity. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

Sharing is defined by coordinates rather than physical storage. For key matrix $\mathbf{K}$ and regional row selector R, $R(\mathbf{K}E_J) = (R\mathbf{K})E_J$:

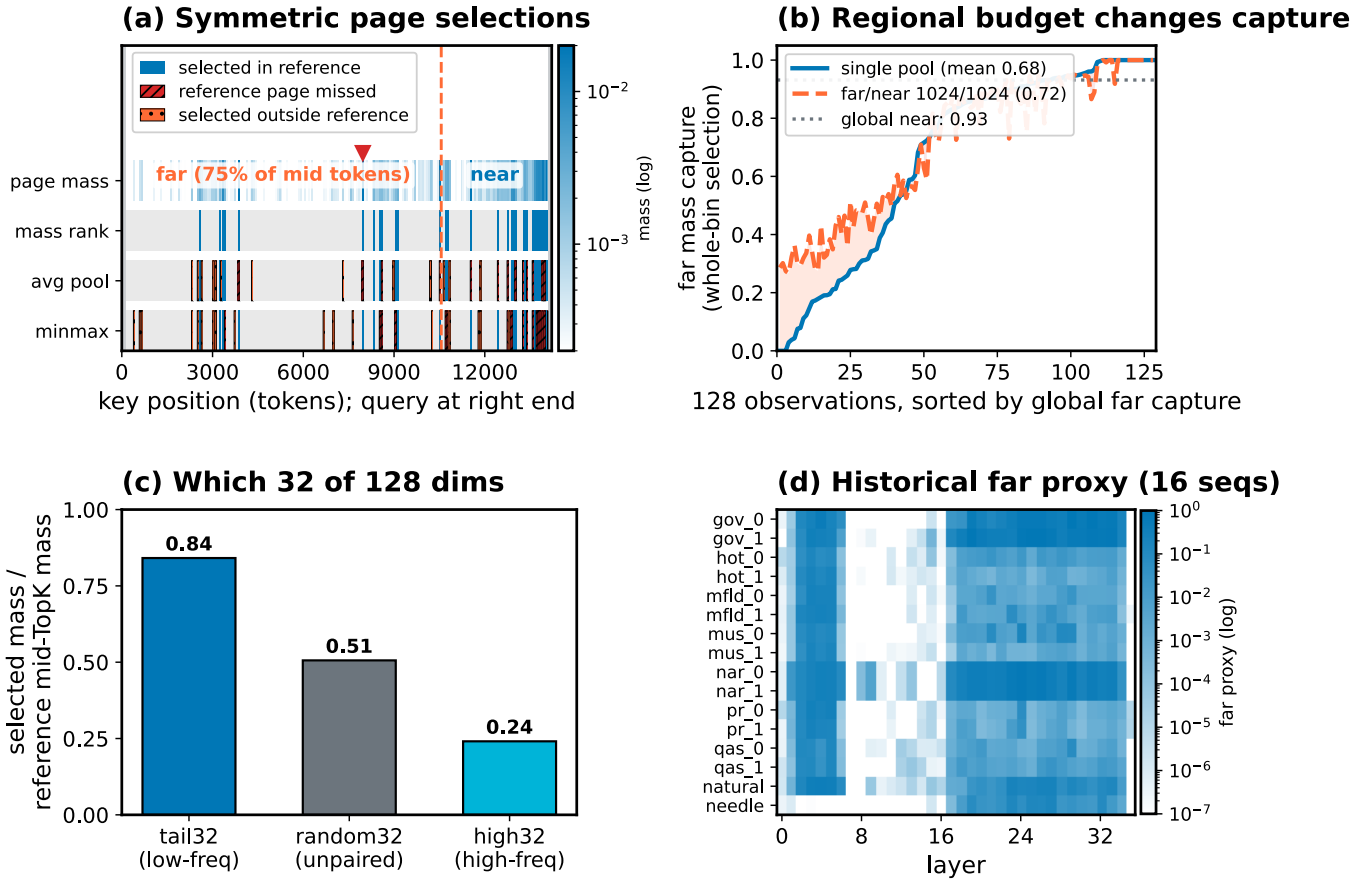


Fig. 1. Four motivating diagnostics; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below are saved mass-rank, avg and minmax sets: 32 pages each, using 1989/1989/2048 tokens under cap2048. Blue denotes selected reference pages; red hatching denotes missed reference pages; orange dots denote selected pages outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Set differences do not establish a quality ranking. (b) Four sequences contribute 128 correlated observations, ordered by global far retention. A global whole-bin cap of 2048 is compared with separate far/near caps of 1024/1024, without refill; red shading marks lower capture. Retention divides selected regional mass by full regional mass; the dotted line is global near retention. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) The same 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the current gate. These are diagnostic selections and profiles, not task-quality or decode-safety results.

the same coordinate map applies before or after partitioning history. L1 derives page summaries and L2 derives quantize–dequantize token keys from these coordinates; the two need not share a buffer. Original keys and values remain full width, so this design does not reduce original-KV cache capacity. Preparing and maintaining the derived representations remain part of total cost; the algebraic identity alone implies no runtime saving (S7).

B. Regional page and token budgets

During decoding, let t be the current token's zero-based position in the full sequence, including prefill tokens, so that $S = t+1$ counts valid cached positions, including the current position and excluding padding. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of the unprotected middle history; far is its complement. Near is therefore distinct from the protected sliding window. The near boundary follows fraction α of unprotected history under minimum-page and alignment rules. The unprotected middle has length $\max(0, S - |\mathcal{P}_t|)$; the selected configuration is specified in the experimental setting.

Figure 2b details the regional cascade. Near uses average-key page scores. Far uses sign-aware extrema in the shared space. For each nonempty region-intersected page b, define $m_{b,j} = \min_{i \in b} z_{i,j}$ and $M_{b,j} = \max_{i \in b} z_{i,j}$. Quest's extrema construction [1] gives

$$U_b(z_q) = \sum_j (z^+_{q,j} M_{b,j} + z^-_{q,j} m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. This bounds unquantized shared-space dot products, not original-dimensional logits or separately quantized scores. Extrema can come from different tokens; a top-page shortlist can still miss important tokens. Average-key scores do not inherit this bound.

With $K_1 = 128$ and $\beta = .125$, near and far receive 16 and 112 L1 pages. Regional reservation allocates access, not a quality guarantee: a far page can displace a useful near page. A later token quota cannot recover a token after L1 discarded its page. This distinguishes regional L1 screening from splitting one global shortlist after selection.

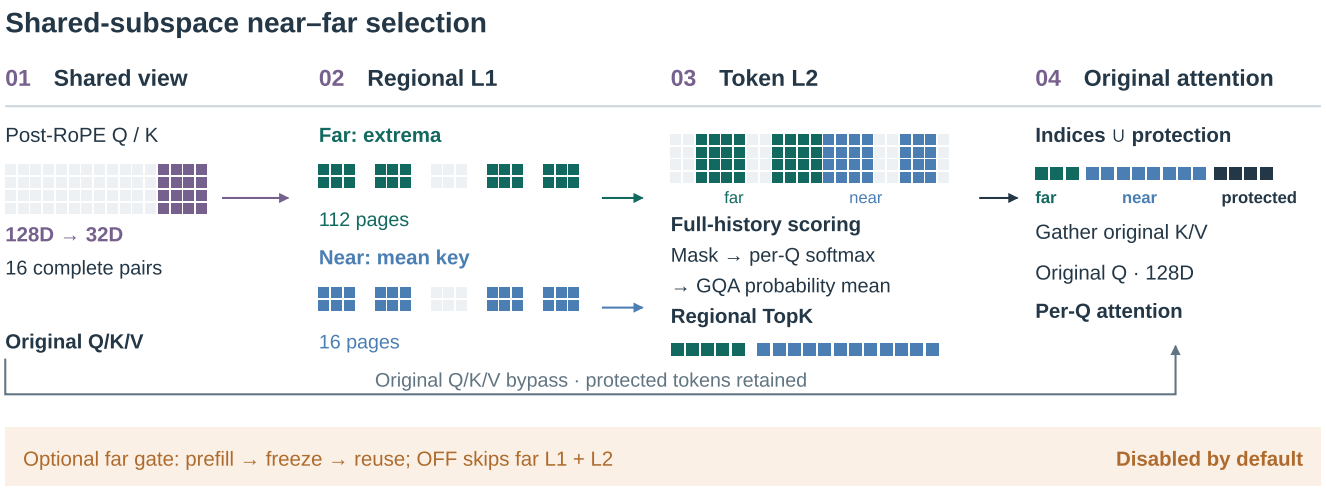


Fig. 2. Shared-subspace selection and the original-Q/K/V bypass. L1 reserves regional pages; L2 masks full-history scores, normalizes each query over shared retained support, averages probabilities within a KV group, and selects regional tokens. When OFF, the optional gate skips far scoring and redirects unused L2 quota to near candidates. Counts illustrate the sufficient-capacity configuration.

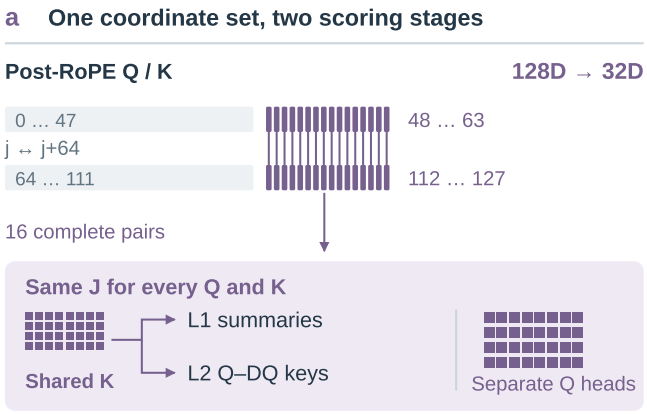


Fig. 2a. Sixteen complete split-half RoPE pairs form the 32-coordinate scoring view. Each layer and KV head derives L1 page summaries and quantize–dequantize L2 keys from the same coordinates; query views remain separate. Original Q/K/V bypass these derived views (Fig. 2).

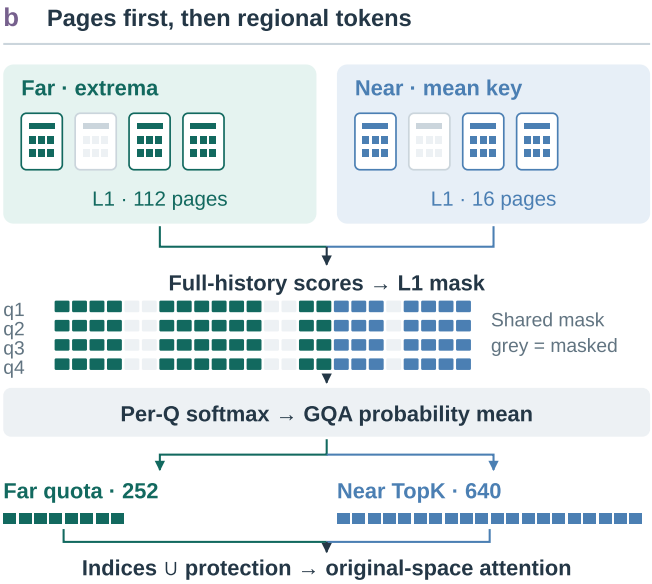


Fig. 2b. Regional screening with full-history L2 scoring. Colored cells indicate eligible candidates, not measured scores. With sufficient eligible capacity, the selected 4/128 prefix/recent configuration has 112/16 far/near pages and nominal 252/640 regional slots, plus 132 protected tokens under cap $K = 1024$. Subpage prefix reselection can consume up to four far slots. Near excludes the protected recent window.

For each KV head h, the intended regional support consists of valid tokens in retained pages. The example below assumes this support can fill each quota. For the supported budget domain $K \geq |\mathcal{P}_t|$, the middle cap is $K_{\text{mid}} = K - |\mathcal{P}_t|$. In the sufficient-length configuration derived from the implementation, a per-head total cap $K = 1024$ leaves 892 nominal middle-token slots (Supplementary Fig. S2). In the standard configuration with sufficient eligible candidates, $\gamma = .625$ acts on near-page capacity, yielding $16 \cdot 64 \cdot .625 = 640$ near tokens and 252 far slots. The selected configuration protects four prefix and 128 recent tokens. Because four prefix tokens occupy only part of a page, far selection can reselect up to four protected positions; 252 is a nominal quota, not a guaranteed count of unprotected far tokens. Scarce-support and zero-tie behavior are detailed in Supplementary Note S7.

C. GQA selection and original-dimensional attention

With grouped-query attention (GQA), 32 query heads and 8 KV heads give four query heads per KV group. L1 averages their page scores before regional TopK. Ordinary L2 scores the full history for each query, applies the shared L1 mask, and averages the independently normalized query probabilities within each KV group. In this Qwen3 implementation, L2 logits retain the original-head scale $1/\sqrt{128}$ before per-query softmax, including when scoring uses 32 coordinates. This probability aggregation differs from applying softmax to averaged logits. Sharing coordinates therefore preserves query-specific distributions, while full-history scoring means that a smaller page shortlist does not itself reduce L2 dot-product work.

The group's unique selected set $\mathcal{S}_{t,h}$ combines protection and regional winners under the total cap. All queries g in that group use these indices, but recompute their own full-dimensional attention, with head width $d = 128$:

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Merging token indices before this softmax avoids adding separately normalized regional outputs [8]. Retained logits use original KV, but restricting the denominator can change attention output. Selected token indices define a common original-space attention set. Physical gathering and reuse depend on the attention backend and sequence length; the inspected implementation also has a short-sequence dense fallback.

D. Prefill-informed far gating

The optional prefill-informed gate (Fig. 2) asks whether prefill observations can identify layers where repeated far selection can be omitted. Per input and layer, mean prefill far-attention weight sum below ϵ sets OFF; otherwise far selection stays active. The decision freezes at decode entry and is reused. The feature is disabled by default. In the supported fixed-quota implementation, OFF skips both far scoring stages and their selections, while unused L2 quota refills near candidates without expanding the near-page budget. Representation preparation and sequence-wide score buffers and normalization remain.

Figure 2c shows the gate decision and refill. We implement this extension for a single sequence; Supplementary Note S5 gives the observation boundary and state transitions. Its threshold controls a prefill statistic rather than subsequent attention error. The gate is therefore an approximation policy whose effect on quality and total cost must be evaluated separately from the default ungated method.

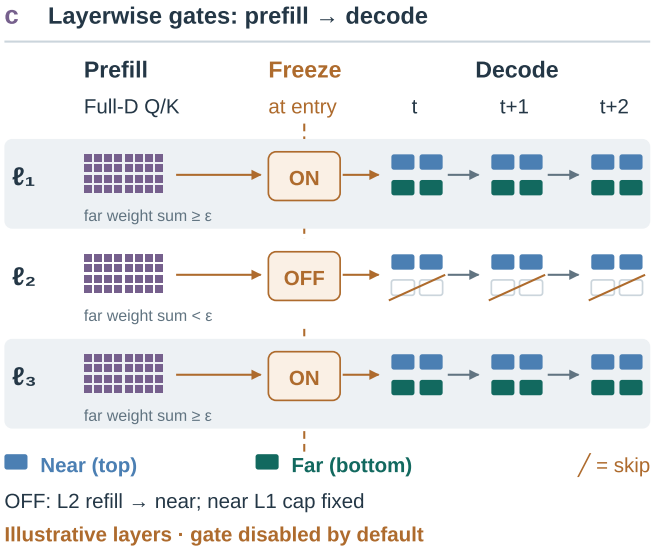


Fig. 2c. Optional far-only control, disabled by default. Full-dimensional causal prefill far-attention weight sum, averaged over the last ≤ 256 query rows and all query heads, sets one decision per input and layer. Decode entry freezes this state; new inputs reset it. OFF skips far L1/L2 and refills near L2 within unchanged near pages. Layer states are illustrative; paired cells denote L1/L2. This statistic does not bound decoding error.

TABLE I
Selected candidate settings: M denotes a configuration parameter, not a completed measurement; C denotes source-derived behavior. K counts unique final tokens including protection.

Parameter	Setting / provenance
Final token cap K	1024 (M)
Page size; L1 page cap K_1	64; 128 (M)
α, β, γ	.25, .125, .625 (M)
L1/L2 active coordinates	32/32 (C)
Prefix / recent protection	4/128 (M)
Prefill; decode layers	dense (M); all sparse (C)
L2 scoring support	full history before mask (C)

IV. Experimental Setting

We separate the design configuration from the configurations that produced the available measurements. Table I specifies the compact Qwen3-8B candidate: dense prefill, sparse decoding and a final unique-token cap of 1024 including four prefix and 128 recent tokens. The five-task quality record instead uses 128/128 protection and 32 active coordinates in zero-filled width-128 L2 keys. The synthetic selector study and model-level timing study are separate experiments. Their outcomes are reported within those scopes; none establishes a matched quality–cost result for the compact candidate. S6/S7 provide the detailed configurations and provenance.

Workloads and comparisons. The quality evidence comprises five LongBench tasks and a historical 11-task RULER [10] 128K comparison. The LongBench scores compare three nominal sparse scoring configurations; dense FullKV is a separate reference in the RULER and decode-timing records. These are not a completed factorial ablation of page and token scoring. A sparse comparison must align final unique-token cap K, including protection, and model, inputs and phase policy. Internal page budgets need not match across methods, but realized token counts and head-sharing granularity affect both access and KV traffic (S7).

Metrics and interpretation. We report task-score means, synthetic selector time and synchronized model-level TPOT separately. Complete-request cost additionally includes preparation, maintenance and input/output work; the archived TPOT diagnostic is not that metric. No task-quality/latency frontier is estimated from unpaired studies. Task-level variation across five workloads is descriptive, not document-paired non-inferiority evidence. Establishing such a claim would require matched runs and a prespecified acceptable-loss margin; the required controls are collected in S7.

V. Evaluation and Discussion

A. Q1: What do archived scoring configurations show?

Descriptive quality comparison. Across HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9], the mean scores are 47.588 (shared-subspace configuration), 47.910 (nominal full-dimensional page-scoring variant) and 47.666 (nominal full-dimensional token-scoring variant). Relative to the shared-subspace configuration, the nominal page- and token-scoring variants are higher by 0.322 and 0.078 points; the latter is 0.244 points below the former. Neither variant is verified as a controlled change in scoring dimensionality. Their score differences are therefore descriptive and do not isolate either stage's effect (Supplementary Fig. S4 and Note S7). The archived shared-subspace configuration score does not directly validate the current compact implementation. None is dense FullKV.

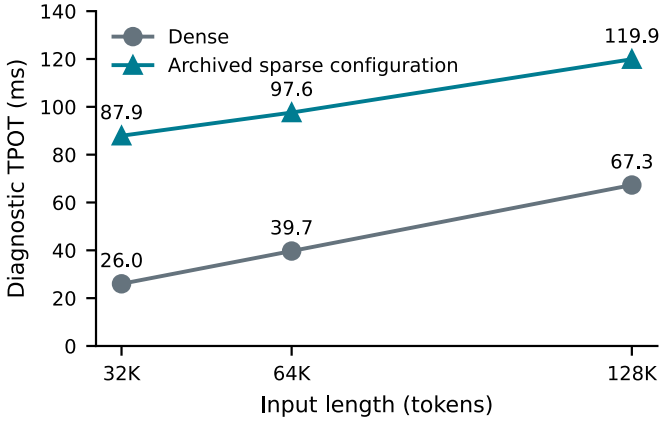


Fig. 3. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

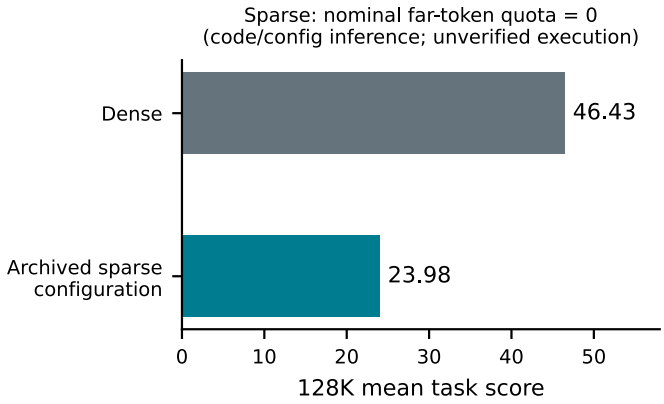


Fig. 4. Author-reported RULER 128K scores: 11 tasks, 100 examples per task and arm. The archived sparse configuration scores 22.45 points below dense FullKV; their macro-average task-score ratio is 51.6%. Reported $\beta = .375$, $K_1 = 128$ and 128/128 protection imply a nominal far-token quota of zero under the inspected formula (middle budget 768; S6). This is a code/configuration inference, not verified execution or causal attribution. The historical setting does not validate the selected 4/128 candidate or support sample-paired inference.

B. Q2: How do regional budgets constrain candidate access?

The regional diagnostic in Figure 1 motivates reserved capacity, but the available task scores do not isolate its benefit. The current compact candidate lacks a matched dense FullKV comparison. Figure 4 preserves the historical 128K scores. Applying the inspected quota formula to the reported $\beta = .375$, $K_1 = 128$, page size 64, $\gamma = .625$ and 128/128 protection gives a nominal far-token quota of zero within the 768-token middle budget. Thus reserving far pages does not by itself guarantee a nonzero final far-token quota. This code/configuration inference does not verify the executed path or isolate the cause of the score gap.

The protection-only configuration scores 26.09 against a 47.91 reference, while using 256 rather than 1024 tokens. The difference therefore combines retrieval-policy and budget effects.

C. Q3: How do reported selection and decode costs compare?

Model-level decode diagnostic. Using dense attention at each input length as the reference, Figure 3 compares author-reported dynamic-cache TPOT for the archived sparse configuration. At 32K, 64K and 128K, sparse medians are 87.9, 97.6 and 119.9 ms, versus 26.0, 39.7 and 67.3 ms for dense attention: sparse-to-dense ratios of reported medians of 3.38, 2.46 and 1.78. Across these three reported lengths, the ratio decreases while sparse TPOT rises and remains above dense TPOT. This synchronized diagnostic excludes tokenization, input transfer and text output; it is not complete-request latency. These aggregate medians do not identify a causal bottleneck; they are not matched quality–cost measurements of the compact 4/128 candidate. Repetition and execution-path limitations are detailed in S6.

Local selector cost. A separate synthetic experiment uses single-layer, batch-one, single-query QKV at length 64K. Its steady-state timing summaries from two rounds are 1.3502 ms for nominal full-dimensional page scoring, 1.3598 ms for shared-subspace scoring and 1.4634 ms for nominal full-dimensional token scoring. The shared-subspace selector thus has no lower reported time than the nominal page-scoring variant in this setting. This ordering does not identify a limiting stage, and the unresolved sampling-protocol discrepancy prevents a reliable interpretation of small timing differences (Supplementary Note S6). The operator timings are not paired with the LongBench quality runs and do not measure complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Prefill statistics also fail to bound later far-region demand: an observed input–layer pair with prefill far-attention weight sum below .05 reaches .1678 during decoding. Relative to the reported reference score of 47.910, gate thresholds of .05 and .10 are associated with mean decreases of .034 and .218 points, respectively; three of five tasks decline in each configuration, including HotpotQA decreases of 1.05 and 1.26 points. The larger per-task losses demonstrate that a small average change can conceal task sensitivity.

The complete scoring configuration is not identified in these gate summaries, so the results cannot be attributed to the gated shared-subspace method (Supplementary Note S6).

Implications. Shared coordinates reduce scoring dimensionality, and regional quotas reserve candidate access; neither establishes a quality–cost advantage. Historical scores do not isolate stage effects, regional comparisons do not establish allocation benefits, and separate timing records cannot establish a matched gain. The optional gate adds approximation and task sensitivity. Non-inferiority and complete-request acceleration therefore require matched evaluation of the compact candidate.

VI. Conclusion

We presented a near–far cascade combining shared low-frequency RoPE scoring and regional page quotas, with original-dimensional final attention. The trade-off is candidate access versus selection cost at a fixed token budget. Historical Qwen3 evidence does not validate the compact candidate's joint quality and cost: synthetic selector timing does not improve on nominal full-dimensional page scoring, sparse TPOT exceeds dense, and prefill statistics do not guarantee decode behavior. The design requires matched, preparation-inclusive evaluation.

Content Generated by AI

OpenAI language models accessed through Codex assisted manuscript drafting and revision, Chinese translation, figure and LaTeX code generation throughout, source checking and arithmetic. The interface did not expose model-version identifiers.

References

[1] J. Tang, Y. Zhao, K. Zhu, G. Xiao, B. Kasikci, and S. Han, "Quest: Query-Aware Sparsity for Efficient Long-Context LLM Inference," in *ICML*, 2024. https://arxiv.org/abs/2406.10774.

[2] L. Ribar, I. Chelombiev, L. Hudlass-Galley, C. Blake, C. Luschi, and D. Orr, "SparQ Attention: Bandwidth-Efficient LLM Inference," in *ICML*, 2024, pp. 42558–42583. https://proceedings.mlr.press/v235/ribar24a.html.

[3] Y. Xu *et al.*, "HISA: Efficient Hierarchical Indexing for Fine-Grained Sparse Attention," in *COLM*, 2026. https://arxiv.org/abs/2603.28458.

[4] Hu *et al.*, "AsyncTLS: Efficient Generative LLM Inference with Asynchronous Two-level Sparse Attention," arXiv:2604.07815v1, Apr. 9, 2026. https://arxiv.org/abs/2604.07815v1.

[5] H. Sun *et al.*, "ShadowKV: KV Cache in Shadows for High-Throughput Long-Context LLM Inference," in *ICML*, 2025, pp. 57355–57373. https://proceedings.mlr.press/v267/sun25b.html.

[6] Z. Wang et al., "MILLION: MasterIng Long-Context LLM Inference Via Outlier-Immunized KV Product QuaNtization," in *2025 62nd ACM/IEEE Design Automation Conference (DAC)*, 2025, pp. 1–7, doi: 10.1109/DAC63849.2025.11132862.

[7] G. Liu, C. Li, J. Zhao, C. Zhang, and M. Guo, "ClusterKV: Manipulating LLM KV Cache in Semantic Space for Recallable Compression," in *DAC*, 2025. https://doi.org/10.1109/DAC63849.2025.11132479.

[8] G. Shen, J. Zhao, Q. Chen, J. Leng, C. Li, and M. Guo, "SALO: An Efficient Spatial Accelerator Enabling Hybrid Sparse Attention Mechanisms for Long Sequences," in *DAC*, 2022, pp. 571–576. https://doi.org/10.1145/3489517.3530504.

[9] Y. Bai *et al.*, "LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding," in *ACL*, 2024, pp. 3119–3137. https://aclanthology.org/2024.acl-long.172/.

[10] C.-P. Hsieh *et al.*, "RULER: What's the Real Context Size of Your Long-Context Language Models?," in *COLM*, 2024. https://arxiv.org/abs/2404.06654.

[11] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient Streaming Language Models with Attention Sinks," in *ICLR*, 2024. https://arxiv.org/abs/2309.17453v4.